

NTUEE Digital Circuit Lab

LAB 1: Random Number Generator Report

Team 5

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1 File Structure

```
teamXX_lab1/  
|-- teamXX_lab1_report.pdf  
+-- NEXYS_A7/  
    |-- NEXYS_A7.xpr  
    |-- NEXYS_A7.srscs/  
    |   |-- sources_1/new/  
    |   |   |-- Top.sv  
    |   |   |-- NEXYS_A7.sv  
    |   |   |-- Debounce.sv  
    |   |   \-- Seven_Segment_Display.sv  
    |   \-- constrs_1/imports/  
    |       \-- Nexys-A7-100T-Master.xdc  
    \-- ...
```

2 System Architecture & Data Path

In our Top.sv, we use a Linear Feedback Shift Register to generate the random number and three variables to record the time to sample the random number as shown in fig.1. The details of the three variable and the FSM are explained in section 3.

3 Hardware Scheduling: Finite State Machine

We designed a three-stage finite state machine. The states include *S_IDLE*, *S_RUN* and *S_DONE* as indicated in fig.2.

- *S_IDLE*: The starting state of our system. The system starts at this state and wait for the BTNU to be pressed. When BTNU is pressed, the system jumps to the *S_RUN* state.
- *S_RUN*: The state that handles random number sampling. After the BTNU is pressed, the system jumps from *S_IDLE* state to this state. The system will sample the random number generated by the LFSR after a certain interval of clock tick. We have two constants: *ONE_COUNT_CYCLE*, which is equal to 1M to indicate the smallest number of clock ticks as the interval of sampling time, and *LAST_UPDATE*, which is equal to 31 to indicate the times of sampling before the sampling process is done. We have three variables to count and increase the interval.

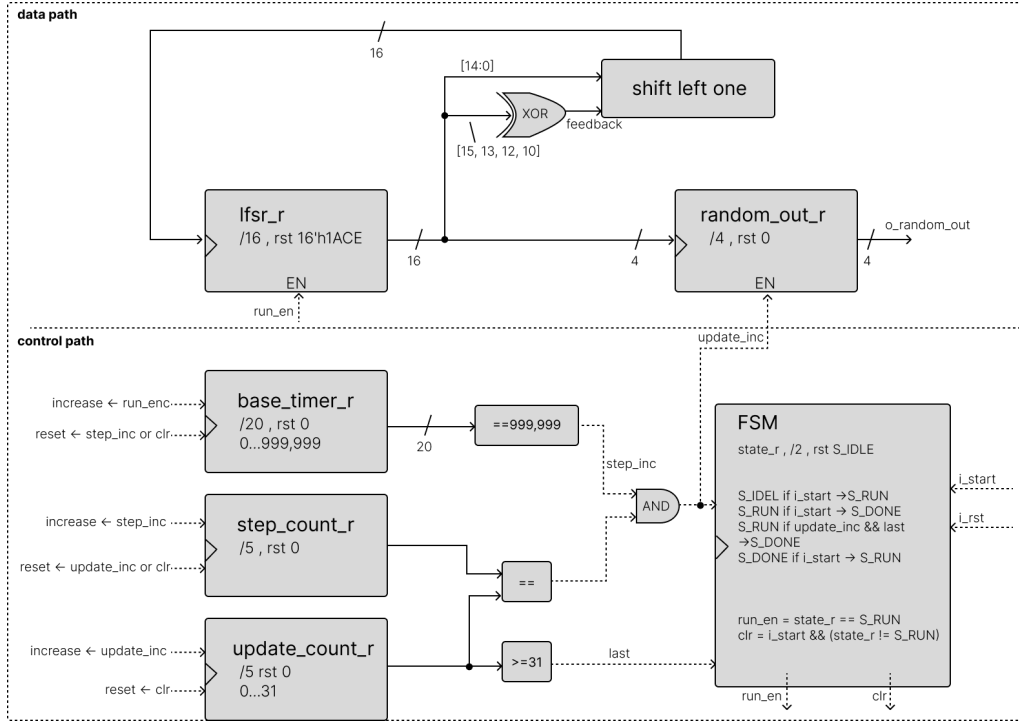


Figure 1: System Architecture and Data Path Diagram

- **base_timer**: Used to record if ONE.COUNT_CYCLE times of clock ticks have passed. If so, the system knows one count cycle has passed, increases step_count by one and initializes base_timer to 0.
- **step_count**: Used to record how many count cycles have passed. It'll be increased until it is equal to update_count, when an interval of sampling is done and the random number generated by the LFSR at that clock tick will be shown. When it reaches update_count, update_count will be increased by one and step_count will be initialized to 0.
- **update_count**: Used to record how many samplings have been done. When it reaches the number of LAST_UPDATE, the system knows the whole sampling process is done and jumps to the S_DONE state.

This state also checks if the BTNU is pressed again during sampling. If so, it'll stop the sampling process, show the random number at that clock tick and jump to the S_DONE state.

- **S_DONE**: The state when sampling is done or when the BTNU is pressed again in S_RUN. The system will jump back to S_RUN if the BTNU is pressed or jump to S_IDLE when the BTNC is pressed.

4 Bonus Features

The bonus feature we implemented is that when the user presses the BTNU again during the sampling process, the system will jump to S_DONE state and show the random number generated at that clock tick. The design method is explained in section3.

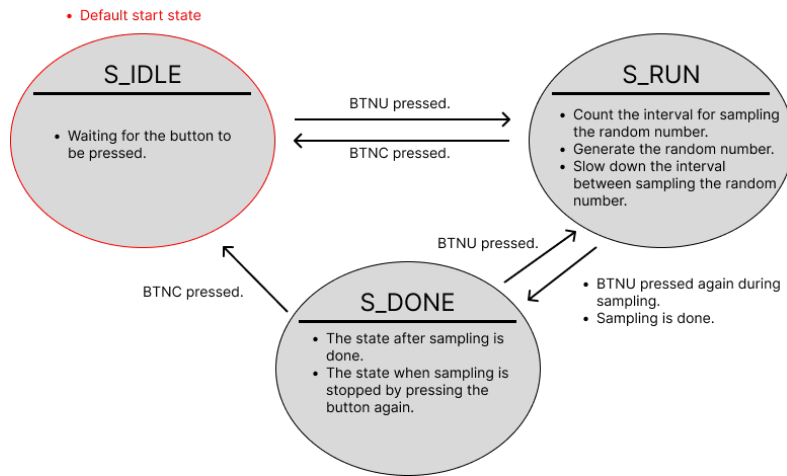


Figure 2: Finite State Machine Transition Diagram

Design Timing Summary					
Setup		Hold		Pulse Width	
Worst Negative Slack (WNS): 5.115 ns		Worst Hold Slack (WHS): 0.169 ns		Worst Pulse Width Slack (WPWS): 4.500 ns	
Total Negative Slack (TNS): 0.000 ns		Total Hold Slack (THS): 0.000 ns		Total Pulse Width Negative Slack (TPWS): 0.000 ns	
Number of Failing Endpoints: 0		Number of Failing Endpoints: 0		Number of Failing Endpoints: 0	
Total Number of Endpoints: 138		Total Number of Endpoints: 138		Total Number of Endpoints: 89	
All user specified timing constraints are met.					

Figure 3: Screenshot of Report Timing Summary

Name	Slice LUTs (63400)	Slice Registers (126800)	Slice (15850)	LUT as Logic (63400)	Bonded IOB (210)	BUFGCTRL (32)
✓ NEXYS_A7	80	88	31	80	72	1
deb0 (Debounce)	26	22	10	26	0	0
seven0 (Seven_Segment_Display)	16	13	6	16	0	0
top0 (Top)	38	53	17	38	0	0

Figure 4: Screenshot of Report Utilization

5 Problems Encountered & Solutions

- **Problem 1:** Our first design has more than 40 slices.

Solution: We originally used a single variable with 27 bits as a counter. We found out that the 27-bit counter was taking too many bits and changed the counter to three separate variables to serve as a more efficient counter with less number of bits.